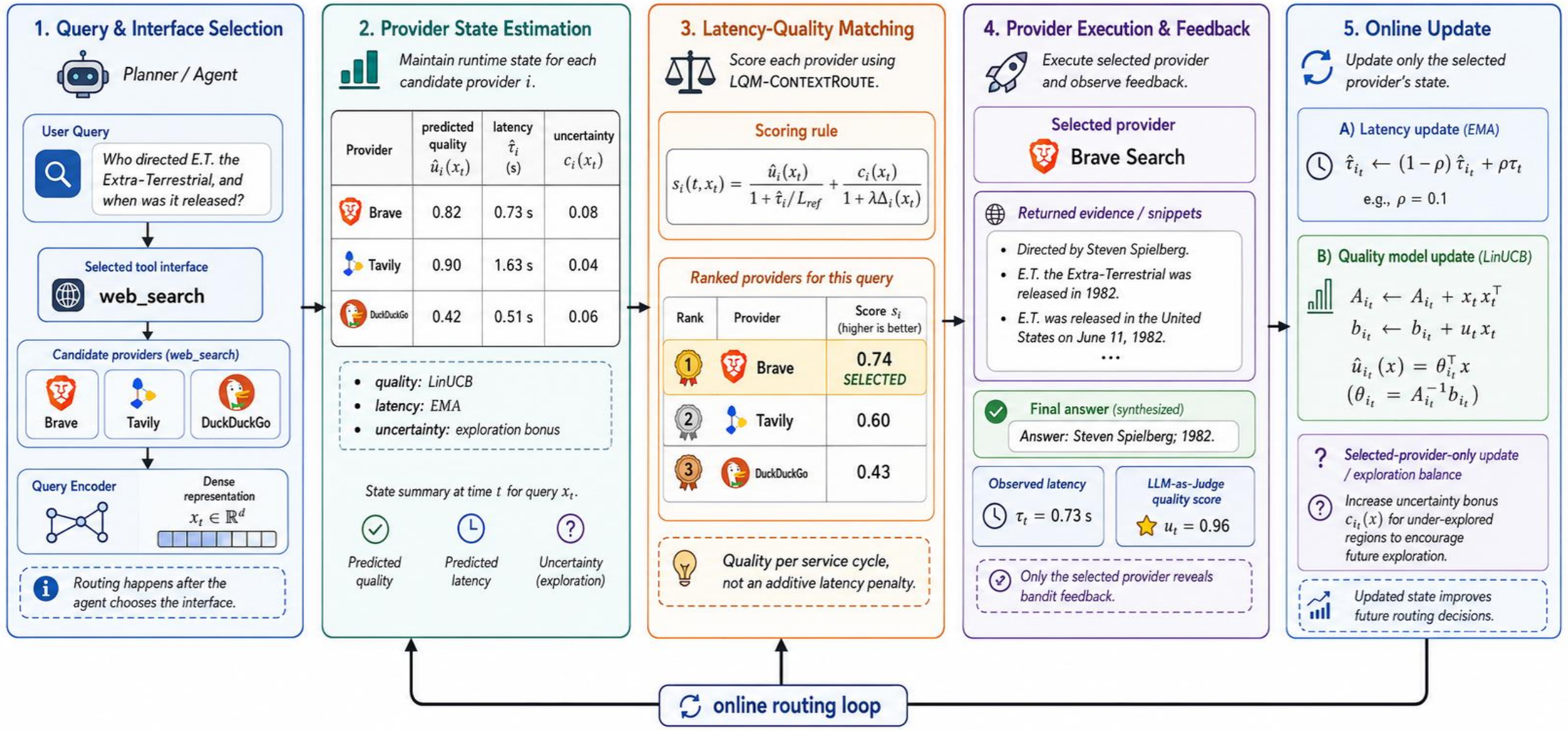

1. Query & Interface Selection
Planner / Agent
User Query
Who directed E.T. the Extra-Terrestrial, and when was it released?
Selected tool interface
web_search
Candidate providers (web_search)
Brave
Tavily
DuckDuckGo
Query Encoder
Dense representation
$x_t \in \mathbb{R}^d$
Routing happens after the agent chooses the interface.
2. Provider State Estimation
Maintain runtime state for each candidate provider i.
Provider
predicted quality $\hat{u}_i(x_t)$
latency $\hat{\tau}_i$ (s)
uncertainty $c_i(x_t)$
Brave 0.82 0.73 s 0.08
Tavily 0.90 1.63 s 0.04
DuckDuckGo 0.42 0.51 s 0.06
quality: LinUCB
latency: EMA
uncertainty: exploration bonus
State summary at time t for query x_t.
Predicted quality
Predicted latency
Uncertainty (exploration)
3. Latency-Quality Matching
Score each provider using LQM-CONTEXTROUTE.
Scoring rule
$s_i(t, x_t) = \frac{\hat{u}_i(x_t)}{1 + \hat{\tau}_i / L_{ref}} + \frac{c_i(x_t)}{1 + \lambda \Delta_i(x_t)}$
Ranked providers for this query
Rank
Provider
Score s_i (higher is better)
1 Brave 0.74 SELECTED
2 Tavily 0.60
3 DuckDuckGo 0.43
Quality per service cycle, not an additive latency penalty.
4. Provider Execution & Feedback
Execute selected provider and observe feedback.
Selected provider
Brave Search
Returned evidence / snippets
Directed by Steven Spielberg.
E.T. the Extra-Terrestrial was released in 1982.
E.T. was released in the United States on June 11, 1982.
...
Final answer (synthesized)
Answer: Steven Spielberg; 1982.
Observed latency
$\tau_t = 0.73$ s
LLM-as-Judge quality score
$u_t = 0.96$
Only the selected provider reveals bandit feedback.
5. Online Update
Update only the selected provider's state.
A) Latency update (EMA)
$\hat{\tau}_{i_t} \leftarrow (1 - \rho)\,\hat{\tau}_{i_t} + \rho \tau_t$
e.g., $\rho = 0.1$
B) Quality model update (LinUCB)
$A_{i_t} \leftarrow A_{i_t} + x_t x_t^\top$
$b_{i_t} \leftarrow b_{i_t} + u_t x_t$
$\hat{u}_{i_t}(x) = \theta_{i_t}^\top x$
$(\theta_{i_t} = A_{i_t}^{-1} b_{i_t})$
Selected-provider-only update / exploration balance
Increase uncertainty bonus $c_{i_t}(x)$ for under-explored regions to encourage future exploration.
Updated state improves future routing decisions.
online routing loop